Friction JEE Assignment

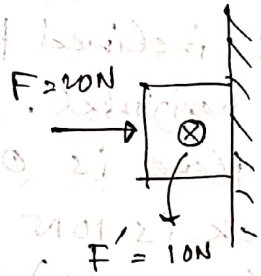
1. A block of mass 'm' is at rest on a rough wedge as shown in fig. What is the force exerted by the wedge on the block

[Ans = mg]

2. A block of mass 1 kg is pushed against a rough vertical wall with a force of 20N, coeff of static friction being $\frac{1}{4}$. Another horizontal force of 10N is applied on the block in a direction parallel to the wall. Will the block move? If yes find its accn?

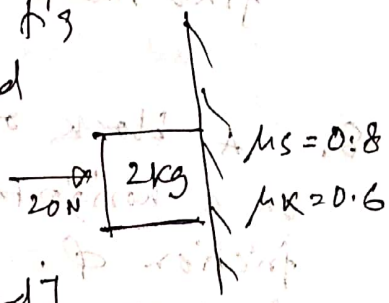
(try later)

[$(2\sqrt{2}-1)$ m/s²]



3. A 2kg block is pressed against a rough wall by a force F = 20N as in fig. Find accn of the block and force of friction acting on it. [g = 10m/s²]

[4m/s² downward, 12N upward]



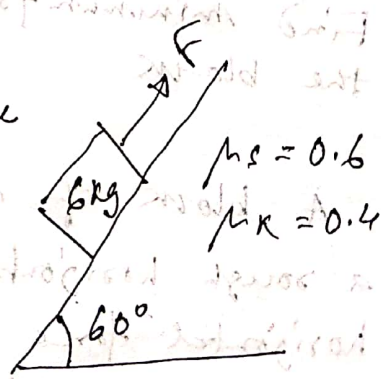
4. A 6kg block is kept on an inclined rough surface as shown in figure. Find the force 'F' required to

a) keep the block stationary,

b) move the block downward with constant velocity

c) to move the block upwards with an accn of 4m/s² (g = 10m/s²)

[a) 34N b) 40N c) 88N]



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5. The time taken by a body to slide down a rough 45° inclined plane is twice that required to slide down a smooth 45° inclined plane. μ_K betn. body and rough plane ? $\left(\mu_K = \frac{3}{4}\right)$ Ans.

6. The force required to just move a body up the inclined plane is double the force required to just prevent the body from sliding down the plane. coefficient of friction is μ. Find $\tan\theta$ if θ = angle of inclination of the plane [3μ] Ans.

7. A block rests on a rough inclined plane making an angle 30° with horizontal. μ_S betn. block and inclined plane is 0.8. If frictional force on the block is 10N then find mass of the block ($g = 10 m/s^2$) [2kg]

8. A block of mass 'm' is placed at rest on a horizontal rough surface with angle of friction ϕ. The block is pulled with a force F at an angle θ with horizontal. Find minimum force 'F' required to move the block $\left[\frac{mg\sin\phi}{\cos(\theta-\phi)}\right]$

9. A block of mass 4 kg is placed on a rough horizontal plane. A time dependent horizontal force $F = Kt$ ($K = 2N/s$) acts on the block. Find friction betn. block and plane at time $t = 2$ sec [$\mu = 0.2$]

[Ans 4N]

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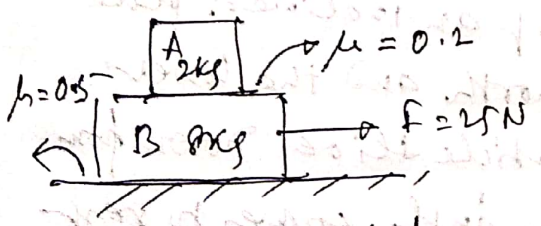
10 A rope of length 'L' and mass 'M' is being pulled on a rough horizontal floor by a constant horizontal force $F = Mg$. The force is acting at one end of the rope in the same direction as the length of the rope. μ_K betn. rope and floor is $1/2$. Find tension at mid pt of rope? [Ans $\frac{Mg}{2}$]

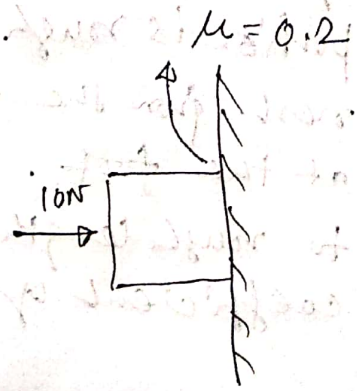
11 A heavy body of mass 25 kg is to be dragged along a horizontal plane [$\mu = \frac{1}{\sqrt{3}}$] Find the least force required for that (125 N)

12 A block A of mass 4 kg is kept on ground. The coeff of friction betn. block and ground is 0.8. An external force 30 N is applied on it parallel to the ground. Find the resultant force exerted by the ground on the block. ($g = 10 m/s^2$) (50 N)

13 A block 'A' of mass 2 kg rests on another block 'B' of mass 8 kg which rests on a horizontal floor. The coefficient of friction betn. A and B is 0.2 & that betn. B and floor is 0.5. When a horizontal force F = 25 N is applied on block B, find the force of friction betn A & B [Ans 0]



14. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. Find weight of the block (2 N)



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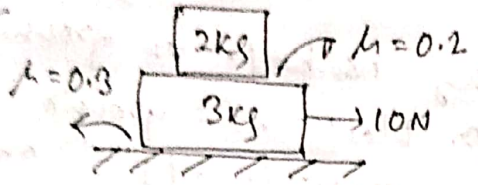
15. The minimum force required to start pushing a body up a rough (coeff of friction = μ) inclined plane is F_1, while the minimum force needed to prevent it from sliding down is F_2. If inclination of the inclined plane be θ, & $\tan\theta = 3\mu$, then find $\frac{F_1}{F_2} = ?$

[Ans 3]

16.



In fig Find friction betn. 3kg block & the plane surface

(Ans 10N)

17. A varying horizontal force $F = at$ [a = +ve const] acts on a block of mass 'm' kept on a smooth horizontal surface. An identical block is kept on the first block. Coeff of friction betn the blocks is μ. Find the time after which the relative sliding betn. the blocks starts ?

$\left[(2\mu mg)/a \right]$

18. The upper portion of an inclined plane of inclination 'α' is smooth and the lower portion is rough. A particle slides down from rest from the top and just comes to rest at the foot. If ratio of smooth length to rough length is m:n, then find the coefficient of friction ?

$\left[\mu = \left(\frac{m+n}{m}\right) \tan\alpha \right]$

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19. A block of mass 0.1kg is held against a wall by applying a horizontal force of 5N on the block. If coeff of friction betn block & wall = 0.5, then find friction acting on the block. [0.1N] {$g = 10 m/s^2$}
Ans.

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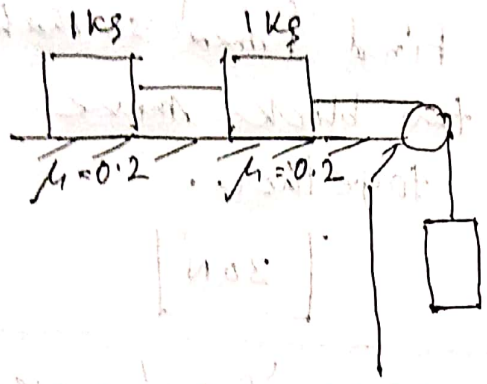
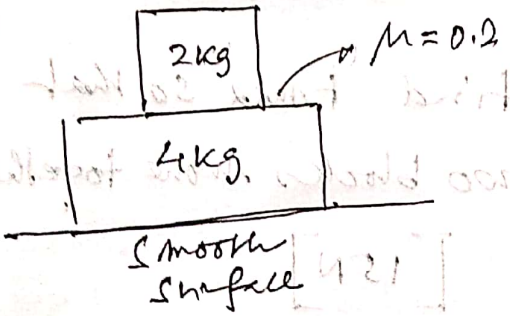


a) Find acelⁿ of 1kg blocks (0.4 m/s²)

b) Find Tension in strings connecting the 1kg blocks. (2.4 N)

c) Find tension in string attached to 0.5kg. (4.8 N)

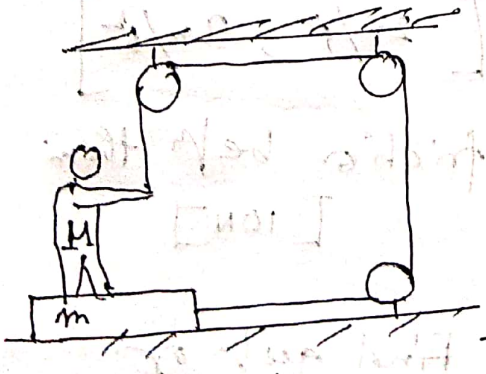
21.



Find acelⁿ of two blocks if a force of 12N is applied to a) upper block b) the lower block [$g = 10 m/s^2$]

[a) upper block 4 m/s², lower block 1 m/s², b) both blocks 2 m/s²]

22.



coeff of friction betn the board and the floor is 'μ'. Find maximum force that the man can exert on the rope, so that the board does not slip on the floor.

$$\left[\frac{\mu(M+m)g}{1+\mu}\right]$$

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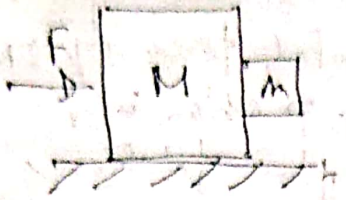
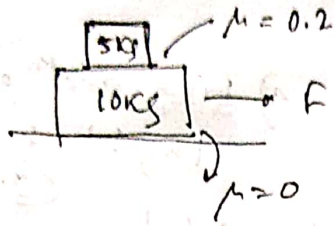


In fig coeff of friction betn M and m is μ. Find minimum horizontal force 'F' required to keep the block $(M+m)\frac{g}{\mu}$.

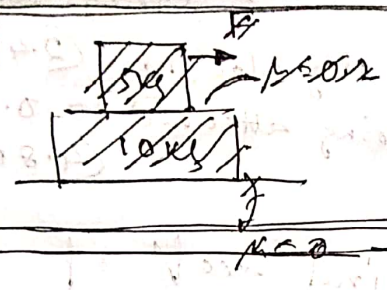
24 a)



Find 'F max' so that two blocks move together.

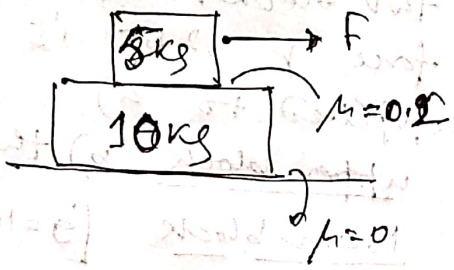
[30N]

~~b)~~



~~Find 'F max' so that~~

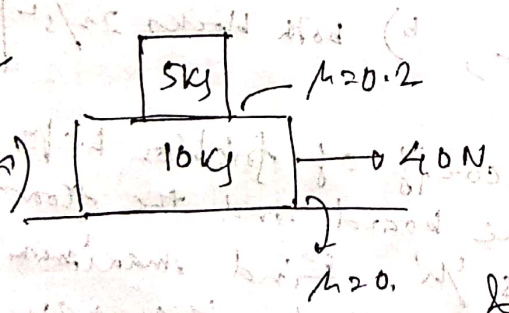
24 b)



Find 'F max' so that two blocks move together

[15N]

25 a)



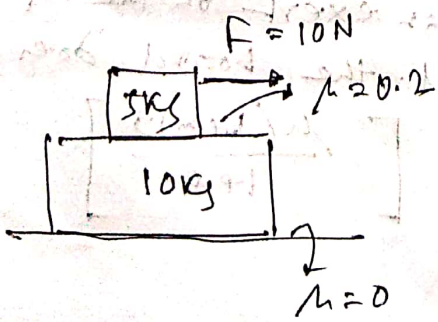
Find accl^n of 5kg & 10kg

[$2 m/s^2$, $3 m/s^2$]

& friction betn them

[10N]

b)



Find accl^n of 5kg & 10kg.
& friction betn them.

$\left(\frac{2}{3} m/s^2, \frac{2}{3} m/s^2, \frac{20}{3} N\right)$

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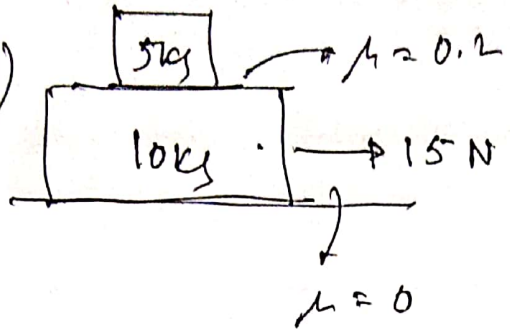
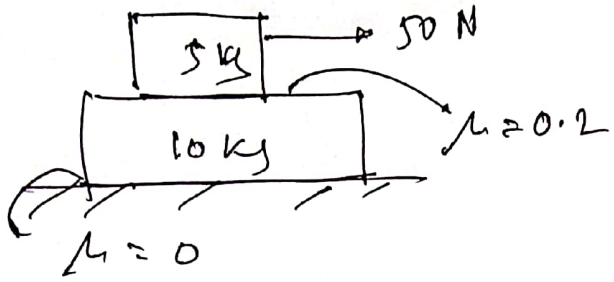
26/ a)



Find acc of 5 kg, 10 kg & friction betn them

[$1 m/s^2$, $1 m/s^2$, 5 N]

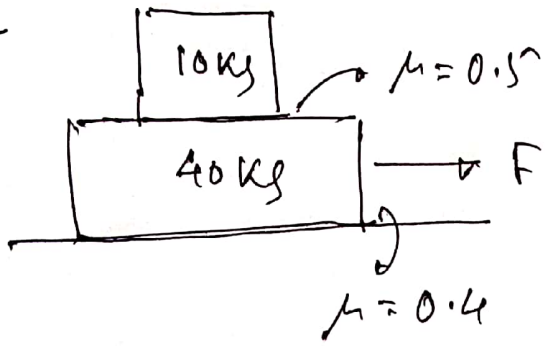
b)



Find acc of 5 kg & 10 kg & friction betn them?

[$8 m/s^2$, $1 m/s^2$, 10 N]

27



Find acc of blocks in fig & friction on blocks if (i) F = 100 N
(ii) F = 300 N
(iii) F = 500 N

Ans [(i) Friction on lower block = 100 N
$a_{10} = a_{40} = 0$, Friction on upper block = 0]

Ans [(ii) $a_{10} = a_{40} = 2 m/s^2$, Friction on 10 kg = 20 N
Friction on 40 kg = 220 N]

[(iii) $a_{10} = 5 m/s^2$, $a_{40} = 6.25 m/s^2$ } Friction on 10 kg = 50 N
Friction on 40 kg = 250 N]

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